

Lejun Min

Researcher, Artist

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EDUCATION

Center for Computer Research in Music and Acoustics, Stanford University

Sept. 2024 – Present

Master of Arts in *Music, Science, and Technology*

California, United States

- GPA: 4.0 / 4.0.
- Advisor: Prof. Marina Bosi, Prof. Takako Fujioka.

Zhiyuan College, Shanghai Jiao Tong University

Sept. 2019 – June 2023

Bachelor of Engineering in *Computer Science* (Fellowship)

Shanghai, China

- Member of **ACM Honor Class**, an elite CS program for **top 5%** students.
- Graduated with **top 1% Outstanding Bachelor's Thesis**.
- GPA: 89/100 (ranking: 5/27).

PUBLICATIONS

L. Min, S. Chen, M. Bosi, “Leveraging Rotational M/S Coding and Machine Learning in Stereo Audio Coding”, in *International Workshop on Sound Signal Processing Applications (IWSSPA 2025)*, Costa Ballena, Spain, July 2025. [[Program](#)]

X. Qu, Y. Bai, Y. Ma, Z. Zhou, K. Lo, J. Liu, R. Yuan, **L. Min**, X. Liu, T. Zhang, X. Du, S. Guo, Y. Liang, Y. Li, S. Wu, J. Zhou, T. Zheng, Z. Ma, F. Han, W. Xue, G. Xia, E. Benetos, X. Yue, C. Lin, X. Tan, S. Huang, W. Chen, J. Fu, G. Zhang, “MuPT: A Generative Symbolic Music Pretrained Transformer”, in *Proc. 13th International Conference on Learning Representations (ICLR 2025)*, Singapore, April 2025. [[arXiv](#)] [[OpenReview](#)] [[Demo](#)]

Z. Wang, **L. Min**, G. Xia, “Whole-song Hierarchical Generation of Symbolic Music Using Cascaded Diffusion Models”, **Spotlight (top 5%)** in *Proc. 12th International Conference on Learning Representations (ICLR 2024)*, Vienna, Austria, May 2024. [[arXiv](#)] [[OpenReview](#)] [[Demo](#)]

L. Min, J. Jiang, G. Xia, J. Zhao, “Polyffusion: A Diffusion Model for Polyphonic Score Generation with Internal and External Controls”, in *Proc. 24th International Society for Music Information Retrieval Conference (ISMIR 2023)*, Milan, Italy, November 2023. [[arXiv](#)] [[Poster](#)] [[Demo](#)]

RESEARCH EXPERIENCE

Smule Labs, Research Intern

Oct. 2025 – Dec. 2025, Mar. 2026 – May 2026

- Built a disentangled music semantic embedding by end-to-end audio captioning with self-supervised learning regularizations. Work under preparation for publication.
- Supervisor: Yongyi Zang.

Sony Computer Science Laboratories - Paris, Research Intern

June 2025 – Sept. 2025

- Designed an end-to-end (re-)mixing and mastering system using audio representation learning and generation. This is a pioneering study on automatic mixing with a fully generative approach. Work under preparation for publication.
- Supervisor: Dr. Stefan Lattner.

SoundPatrol & Stanford University, Research Assistant

Feb. 2025 – May 2025

- Trained a singer representation model that discerns deep fake singing synthesis as potential copyright infringement. Reduced the equal error rate (EER) by 10% and boosted the top-1 accuracy up to 96% on the *SingFake* dataset.
- Advisor: Prof. John Thickstun, Prof. Walter De Brouwer.

Music X Lab, MBZUAI, Research Assistant

Sept. 2023 – Feb. 2024

- Designed and implemented comprehensive experiments for the hierarchical generation of symbolic music, with a cascaded diffusion model as backend. Work published at ICLR 2024.
- Advisor: Prof. Gus Xia.

Music X Lab, MBZUAI, Research Assistant

June 2022 – Dec. 2022

- Achieved state-of-the-art polyphonic music generation using diffusion models, with two novel control paradigms: internal control via masked generation, and external control via cross-attention mechanism. Work published at ISMIR 2023.
- Advisor: Prof. Gus Xia.

SKILLS

Research Specialties	Music Generation, Music Information Retrieval, Representation & Multimodal Learning, Digital Signal Processing, Human-Computer Interaction
Programming Languages	C++, C, Python, Java, Rust, Verilog, Lua, Arduino, WGSL, Bash, LaTeX, Typst
Machine Learning	PyTorch, TensorFlow, Accelerate, Lightning
Development Environment	Arch Linux (main OS), Neovim (main editor), Ubuntu, VSCode, Git, Docker
Audio & Graphics Software	JUCE, ChucK, Reaper, Adobe Audition, FL Studio, Pure Data, Audacity, Blender, Adobe Premiere, Kdenlive, Krita

AWARDS & HONORS

CCRMA Flagship Project Award	Feb. 2025
Chiang Chen Overseas Graduate Fellowship (one of 10 awardees in Mainland China)	Jan. 2025
CCRMA Fellowship	Sept. 2024
SJTU Outstanding Bachelor's Thesis (one of 41 awardees out of 3873 graduates in 2023)	June 2023
Longhu Scholarship (top 5% in Zhiyuan College)	Apr. 2023
SJTU Student of Merit (one awardee in each major)	Dec. 2021
Zhiyuan Honorary Scholarship	2019 - 2023

TEACHING

Perceptual Audio Coding (Music 422), Stanford University , Teaching Assistant	Winter 2026
• Taught a review lecture on perceptual audio codec components. Renovated coding environment setup with a template repository. • Lecturer: Prof. Marina Bosi.	
Reinforcement Learning (CS 3316), SJTU , Teaching Assistant	Spring 2023
• Designed a final project that involves single- or multi-agent learning for simulated hands and legged robot. • Lecturer: Prof. Weinan Zhang.	
Design and Analysis of Algorithms (AI 2615), SJTU , Teaching Assistant	Spring 2022
• Prepared well-written lecture notes and answers for assignments. • Lecturer: Prof. Chihao Zhang.	
Principle and Practice of Computer Algorithms (CS 1952), SJTU , Teaching Assistant	Summer 2021
• Designed a comprehensive ray tracing tutorial written in the Rust language. The repository received 100+ stars on GitHub. • Lecturer: Prof. Yong Yu.	

PROGRAMMING PROJECTS

Computer Graphics

Gigantic Splight (Python)

June 2022

An interactive 3D fluids simulation based on Taichi framework.

Scotty3D (C++)

Mar. 2022

A comprehensive CG project including software rasterization, interactive mesh editing, path tracing, and dynamic animation.

Ray Tracer (Rust)

Aug. 2020

A complete ray tracing engine in Rust.

Compiler & Computer Architecture

Mx Compiler (Java)

May 2021

A completely hand-made compiler for a toy language (Java subset) that surpasses -O1 optimization.

RISC-V CPU (Verilog)

Dec. 2020

An emulated 5-pipelined RISC-V CPU with real-world FPGA implementation.

Python Interpreter (C++)

Feb. 2020

A Python language interpreter.

Algorithm & Data Structure

Train Ticket System (C++)

June 2020

A cooperated project including backend coding, B+ Tree data structure implementation and frontend website design.

ART PRACTICES

Live Performance & Intermedia Art

Sound Poetry, for 2-channel audio & vocalists

Dec. 2025

A musique concrète sound poem about dream, and a mandarin (grape) fugue.

Umbrella, for Ambisonics audio & video

Dec. 2025

An intermedia piece exploring the nature of self and fear.

Interplanetary Concert, for 2-channel audio, video, & live performer

Oct. 2025

Breaking the fourth wall with the interplanetary teleportation system.

A Chan Conversation, for Ambisonics audio, Gametrak, & live performer

May 2025

A sonic conversation with an ancient Chan Buddhist monk. Performed on CCRMA Open House Concert 2025.

Interface & Interactive Design

Sonic Skateboard (Arduino & Chuck)

June 2025

Turn my skateboard into a musical instrument.

Talking to A Black Hole (Chuck & ChuGL & WGSL)

Dec. 2024

Let the noise guide you through the event horizon of a lonely black hole.

Kandinsky Sonified (Chuck & ChuGL)

Nov. 2024

An interactive audiovisual music sequencer that creates and sonifies Kandinsky-like abstract paintings.

Fireflies (Chuck & ChuGL)

Oct. 2024

An interactive music therapy journey embodying a firefly. Essentially a sound peeking visualization.

Music & Sound Art

The Backrooms: Audio Drama, for binaural audio

Dec. 2024

The protagonist “no-clipped” into a weird space where he heard things beyond his comprehension.

忆久 (Memories Last Long)

June 2023

A song and a music video dedicated to the Zhiyuan College graduates of 2023.

Should Have Known Better (piano & synth cover)

Feb. 2023

Piano, synth, & singing recording.

晚海 (Sunset Sea)

Dec. 2021

An electronic music piece published under CEM Records.

OPEN-SOURCE CONTRIBUTIONS

canvas-downloader (*Rust*), Author & Packager

A command-line tool to download and organize all your Canvas course materials.

emailbook-rs (*Rust*), Author & Packager

A minimalistic email address book with aerc integration.

chuckfmt (*Rust*), Author & Packager

A fast code formatter for ChuckK programming language.

cli-toobox (*Shell Scripts*), Author

A curated collection of modern command-line tools and configurations for terminal-based development.

AUR (*Shell Scripts*), Packager

My managed Arch User Repository packages with automated version tracking and updates.

plugdata (*C++*), Packager

A plugin wrapper for Pure Data, featuring a new GUI made with JUCE.

COMMUNITY SERVICE

International Computer Music Conference (ICMC) 2026, Reviewer

2026

International Computer Music Conference (ICMC) 2025, Reviewer

2025

Large Language Models for Music & Audio (LLM4MA) Workshop at ISMIR 2025, Reviewer

2025

ChuckK Development Team, CCRMA, Stanford University, Contributor

2025 - Present

Game Audio Initiative, Stanford University, Founder

2025 - Present

DSP Seminar, CCRMA, Stanford University, Speaker & Contributor

2024 - Present

Arch User Repository (AUR), Maintainer (id: aik2)

2022 - Present

LANGUAGE PROFICIENCY

Mandarin Chinese (native), English (fluent), French (beginner)

TOEFL: 112 (Reading 30, Listening 30, Speaking 24, Writing 28)

GRE: Verbal 162, Quantitative 170, Writing 4.0